



BLKEXPORT User Guide

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Software
Version 1.0

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02. Quick Start: DXF to BOM in 3 easy steps

1. **Open**
File > Open DXF or Ctrl+O. Pick your drawing.
→ Check *Integrity Summary* shows PASS.
2. **Review**
Click the tabs: **Attributed** | **Non-Attributed** | **Anonymous**
→ These are your blocks, auto-sorted.
3. **Export**
Ctrl+E = Full block schedule CSV (blue button)
Ctrl+B = Summarised BOM CSV (green button)

Done. Open the CSV in Excel.

03.Overview

BLKEXPORT is a desktop utility designed to process DXF drawings and identify block insertions. The application categorises blocks into Anonymous Blocks, Non-Attributed Blocks, and Attributed Blocks, then allows the results to be exported to CSV files. It also provides a summarised BOM (Bill of Materials) export and full block schedule export.

04. Who is *BLKEXPORT* for?

BLKEXPORT is designed for anyone who needs to generate a Bill of Materials (BOM) from AutoCAD drawings. It automatically extracts all blocks from a drawing and compiles them into a BOM.

Landscape architects can use BLKEXPORT to count every plant and tree in a landscape design. M&E engineers can generate a list of all electrical components in a building. Process engineers can produce a BOM from P&ID drawings to schedule valves, vessels, pumps, and other equipment.

If your drawings use blocks, BLKEXPORT can generate your BOM automatically.

05. System Requirements & Known Limits

Minimum Requirements

- Operating System: Windows 11 Pro
- Installed RAM 8 GB
- Display 1440×900 resolution
- Storage 50 MB free space
- System type 64-bit operating system, x64-based processor

Tested Limits

These limits reflect the test environment used for v1.0 validation. BLKEXPORT does not impose its own block count limit, but was tested with files created in progeCAD 2021.

Block count: Tested up to 100,000 blocks per DXF file. This upper bound is imposed by progeCAD 2021, which was used to generate the test files. BLKEXPORT successfully processed all files up to this limit. Behaviour above 100,000 blocks is untested.

DXF file size: Successfully tested up to 46 KB for ASCII DXF and 29 KB for Binary DXF containing 100,000 blocks.

Note: File size depends on geometry and attribute complexity. The 46 KB test file used simple blocks with minimal geometry. Drawings with complex blocks or extensive attribute data will be larger and may load more slowly.

Memory usage: Loading a 100,000-block DXF file used approximately 1.8 GB of RAM on the reference test machine. Systems with 4 GB RAM may experience slowdowns or paging to disk when processing files near this limit.

Attribute length: Maximum 255 characters per individual attribute, as defined by the DXF specification. Total attributes per block are limited by available memory.

Supported DXF formats: AutoCAD 2000 through AutoCAD 2018, inclusive. Both ASCII and Binary DXF variants are supported.

Unsupported formats: DWG format is not supported. Convert to DXF first using AutoCAD or progeCAD.

Unsupported block types: Dynamic Blocks and Smart Blocks are not supported. They will be ignored during processing and excluded from counts/exports.

DXF load performance: Loading a DXF file containing 100,000 blocks takes approximately **2.5 minutes** on the reference test machine. During loading, BLKEXPORT may appear unresponsive while the file is being processed. Actual load times depend on host machine performance, DXF complexity, and whether the file is on a local SSD or network drive. Once the DXF file is loaded, exporting the summarised BOM and full block schedule to CSV is instantaneous.

Reference Test Machine Specification

- Operating System: Windows 11 Pro
- Operating System Version: 25H2
- Processor 13th Gen Intel(R) Core(TM) i9-13900 (2.00 GHz)
- Installed RAM 32.0 GB (31.6 GB usable)
- Graphics card Intel(R) UHD Graphics 770 (128 MB)
- Storage 1TB SSD
- Display 2560 × 1440 resolution
- System type 64-bit operating system, x64-based processor

Notes

- If *Integrity Summary* shows FAIL, check file format first.
- Network drives supported, but local SSD recommended for files >10MB.

06.Supported Files

Supported formats

- AutoCAD 2000 through AutoCAD 2018, inclusive. Both ASCII and Binary DXF variants are supported.
- DWG not supported.

07. Install BLKEXPORT Software

The BLKEXPORT software comes with a built-in installer.

1. **Run the installer**
Right-click setup.exe and choose **Run as administrator**.
2. **Choose install location**
Keep the default path C:\Program Files\BLKEXPORT\ and click **Next**.
3. **Set user access**
Select **Install for Everyone**, then click **Install**.
4. **Finish**
When installation completes, the BLKEXPORT desktop icon will be created.

BLKEXPORT is now installed. Double-click the desktop icon to launch the application.

08. Uninstall BLKEXPORT Software

1. **Launch Control Panel**
Press Win + R, type control, and press Enter.
2. **Open Programs and Features**
Click **Programs > Programs and Features**.
3. **Uninstall BLKEXPORT**
Scroll to **BLKEXPORT**, right-click, and choose **Uninstall**. Click **Yes** to confirm.
4. **Done**
BLKEXPORT is now removed from your computer. The program files and desktop icon are also deleted automatically.

09. Software Interface

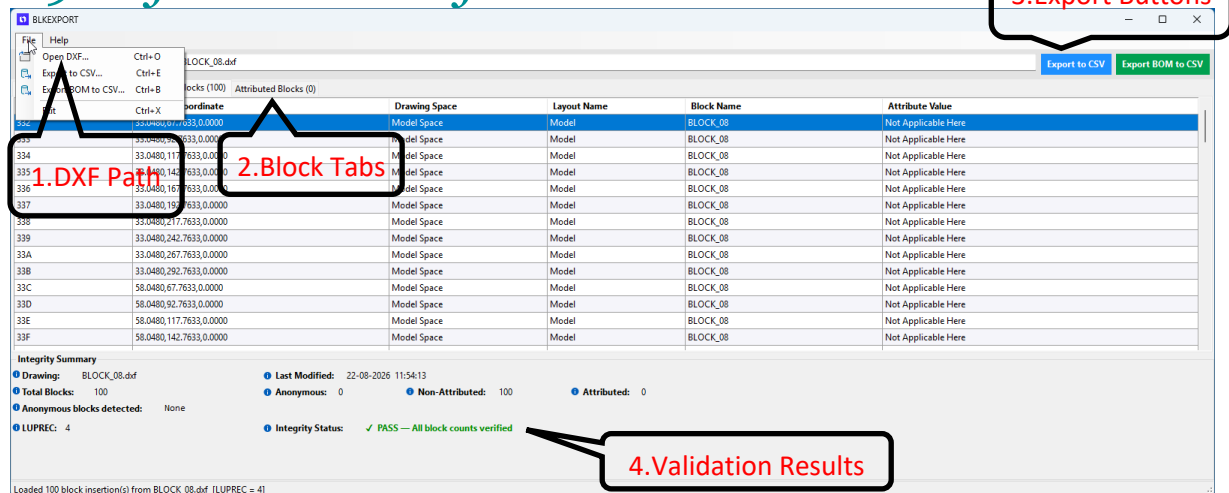


Figure 1 Main application dialog after loading a DXF. Note the 3 block tabs for filtering and colour-coded export buttons — blue for full schedule, green for BOM. Check the Integrity Summary bottom-right to confirm PASS before exporting

Hover your mouse over the information icons for interactive tooltip help.

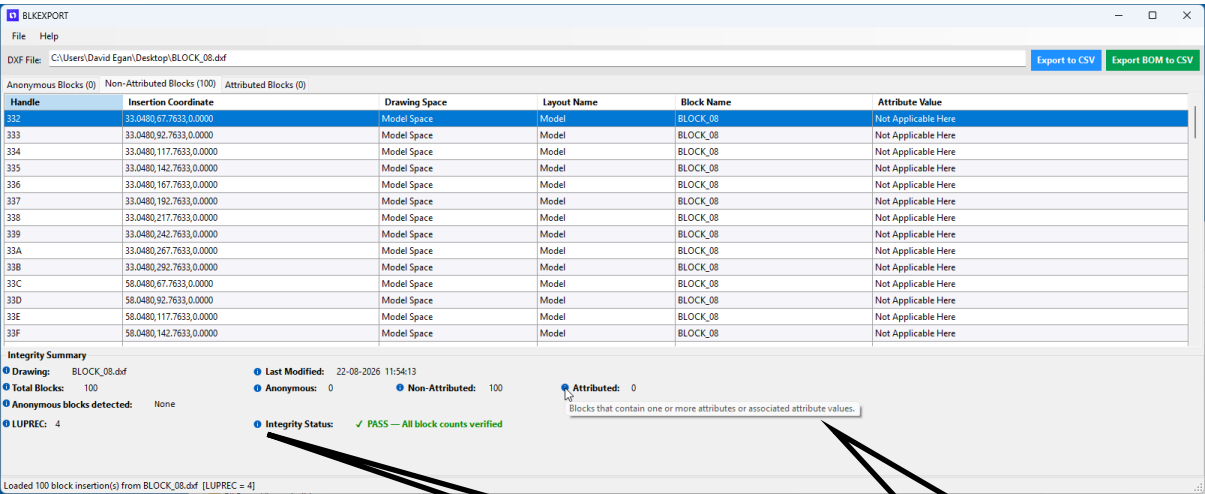


Figure 2 Interactive Tooltip Help

1.Information Icon

2.Interactive tooltip help

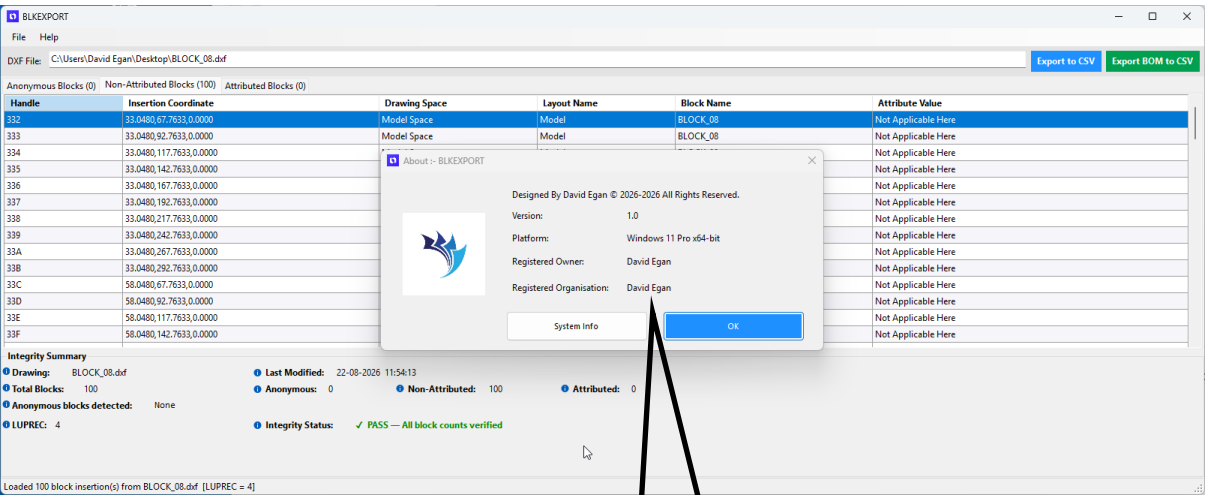


Figure 3 Main application dialog with About dialog

1.About Dialog

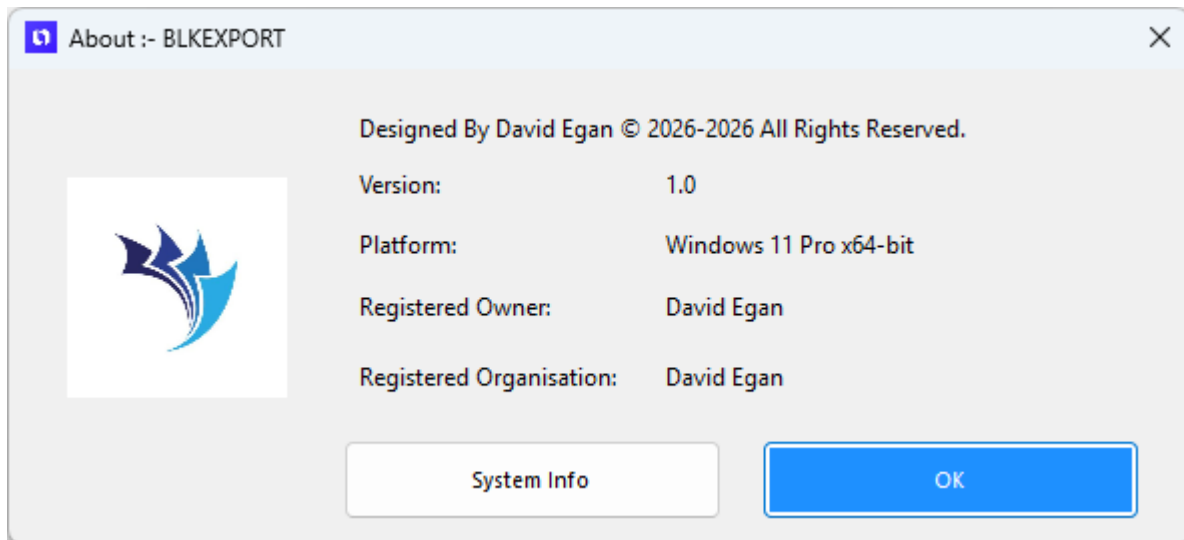


Figure 4 The About dialog displays version information, platform support, operating system registration details, and access to system information

10.Menu Commands

File Menu:

- Open DXF (Ctrl+O) – Open and process a DXF drawing.
- Export to CSV (Ctrl+E) – Export full block schedule data blue button.
- Export BOM to CSV (Ctrl+B) – Export a summarised BOM (Bill of Materials) green button.
- Exit (Ctrl+X) – Close the application.

Help Menu:

- About – Displays application version, registration information and system information.

11.Loading a DXF File

1. Select File > Open DXF or press Ctrl+O.
2. Browse to the required DXF file.
3. The application processes the drawing and populates the block tables.
4. Review the Integrity Summary to confirm successful processing.

12. Block Categories

Anonymous Blocks – Displays anonymous block references detected in the drawing.

Non-Attributed Blocks – Lists blocks without attributes.

Attributed Blocks – Lists blocks containing attribute values.

13. Data Columns

Typical columns include:

- Handle – Unique entity handle.
- Insertion Coordinate – Block insertion point.
- Drawing Space – Model Space or Paper Space.
- Layout Name – Associated layout.
- Block Name – Name of the inserted block.
- Attribute Value – Attribute content when applicable.

14. Exporting Data

Export to CSV:

1. Load a DXF file.
2. Click Export to CSV blue button or press Ctrl+E.
3. Choose a save location.
4. Open the generated full block schedule CSV in Excel or another spreadsheet application.

Export BOM to CSV:

1. Click Export BOM to CSV green button or press Ctrl+B.
2. Select a destination folder.
3. Open the generated summarised BOM (Bill of Materials) CSV in Excel or another spreadsheet application.

15. Integrity Summary

The Integrity Summary section provides:

- Drawing name (dxf) and modification date.
- Total block count.
- Anonymous, attributed, and non-attributed block counts.
- LUPREC value used for coordinate precision.
- PASS/FAIL status confirming that detected block counts have been verified.

16. Software Design Philosophy

No AI. No fluff. Just engineering.

BLKEXPORT is built like the structures I trust — with reinforced concrete and steel.

What that means:

- **Predictable:** Does exactly what you expect, every time you run it.
- **Dependable:** No hidden calls home. No random updates breaking your workflow.
- **Built to last:** Engineered for real CAD work, not demos or trends.

I build software the way it was 20 years ago: solve the actual problem, then get out of your way.

Like a good bridge, it should work on day 1 and day 1001. Same result. No surprises.

That's the standard I hold myself to.

That's BLKEXPORT.

David Egan

Sole Software Developer, System Designer & Solutions Architect

17. Troubleshooting

No data displayed:

- Verify the selected file is a valid DXF.

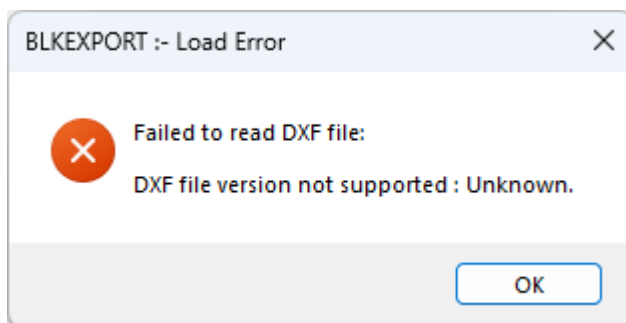


Figure 5 Invalid DXF file selected

- Ensure the drawing contains block insertions.

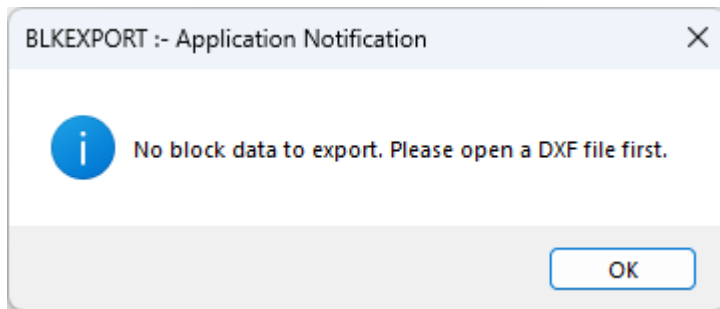


Figure 6 No blocks in the drawing to export, or no drawing selected

Export unavailable:

- Load and process a DXF file first.
- Verify you have write permission to the target folder.

Integrity check failure:

- Re-save the drawing as a supported DXF format and reload it.

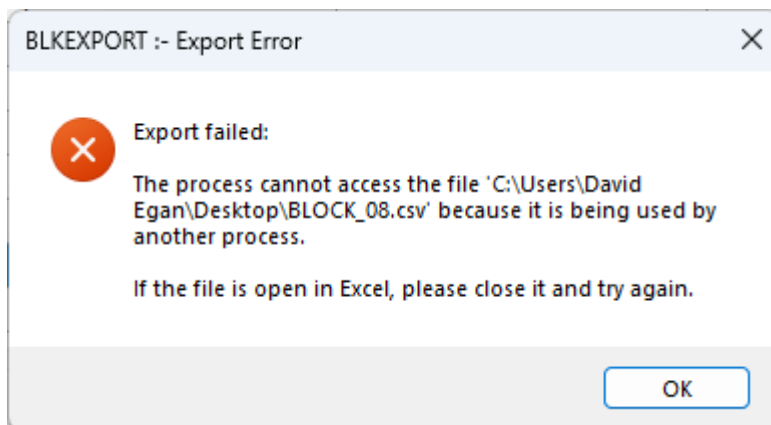


Figure 7 File open in Excel

- The BLOCK_08.csv file is current open in Excel. Close this file in Excel before exporting the CSV file again.

18. Licensing & Registration

Licence type: BLKEXPORT v1.0 is licensed as perpetual, single-user software.

Activation: No internet connection or activation required. The software runs fully offline.

Registration: Registration details are displayed in Help > About For commercial licences, the registered company name appears here. Unregistered copies display “Unregistered”.

Trial limits: There is no time-limited trial. Unregistered copies are fully functional.

No calls home: BLKEXPORT does not contact external servers for licence checks, updates, or telemetry.

If you require a commercial licence or volume licensing, contact: licensing@blkexport.com

19. Updates & Support

Software updates: BLKEXPORT does not auto-update. This is intentional — no random updates breaking your workflow. New versions are announced at www.blkexport.com. To update, download the new installer and run it. It will replace the previous version.

Support policy: BLKEXPORT is provided as-is. For bugs that prevent core functionality — loading a valid DXF or exporting CSV — report them to support@blkexport.com with:

1. BLKEXPORT version from Help > About
2. Windows version
3. The DXF file, if possible
4. Screenshot of the Integrity Summary

Response is on a best-effort basis. Feature requests are logged but not guaranteed.

No phone support: Email only. This keeps the software cost down and your workflow uninterrupted.

20. Keyboard Shortcuts

Ctrl+O – Open DXF file

Ctrl+E – Export to CSV in blue button

Ctrl+B – Export BOM to CSV in green button

Ctrl+X – Exit

21. Glossary

Anonymous Blocks: AutoCAD-generated blocks with names like *U1, *D123. Typically created by dimensions, hatches, or dynamic blocks. BLKEXPORT lists them separately as they cannot be renamed meaningfully.

Attributed Blocks: Blocks that contain attribute definitions for storing text data, e.g. part numbers or descriptions.

BOM: Bill of Materials. A summarised list of block names and quantities.

DXF: Drawing Exchange Format. An ASCII or binary file format for CAD data interchange. BLKEXPORT supports DXF only.

Entity Handle: A unique hexadecimal identifier assigned to every object in an AutoCAD drawing.

LUPREC: AutoCAD system variable controlling the decimal precision displayed for linear units. BLKEXPORT uses this value when rounding block insertion coordinates in exports.

Non-Attributed Blocks: Blocks without attribute definitions. Only the block name and insertion data are exported.

Paper Space / Model Space: AutoCAD drawing contexts. BLKEXPORT reports which space each block insertion belongs to in the full block schedule CSV export.

22. *The Importance of Meaningful Block Names*

Using meaningful block names makes the Summarised BOM and Full Block Schedule CSV exports much easier to read and understand.

After exporting your blocks using either the BOM or Full Blocks Schedule CSV option, you may encounter obscure or meaningless block names in the exported CSV file.

To identify these obscure blocks and determine what they represent, follow the steps below:

1. **Locate the obscure block name**
Review the exported CSV file and identify any block names that are unclear or meaningless.
2. **Copy the insertion coordinate**
Open the Full Block Schedule CSV and locate the obscure block. Copy its insertion coordinate.
3. **Open the drawing**
Open the drawing that contains the obscure block in CAD.
4. **Locate the block using the insertion coordinate**
Invoke the **LINE** command. When prompted for the From point, paste or enter the insertion coordinate copied from the Full Block Schedule.
A line will begin at the insertion point of the obscure block. This provides a visual reference to the exact location of the block in the drawing.
5. **Identify the block**
Use the line's starting point to locate and identify the block in the drawing. Determine what the block represents and what its meaningful name should be.
6. **Rename the block**
Once the block has been identified, use the Rename Block command to assign a meaningful and descriptive name to the block.
7. **Save the modified drawing as DXF**
Save the updated drawing in DXF format.
8. **Re-export the block schedules**
Close the DXF file in CAD. Open the modified DXF in the BLKEXPORT software and generate the relevant CSV files again.
9. **Verify the block name**
Review the newly generated CSV files. The obscure block name that appeared in the previous CSV should now display the meaningful block name you assigned.
10. **Repeat as required**
Repeat this process for each obscure block name identified in the exported CSV files.

Following this procedure ensures that obscure or legacy block names are replaced with meaningful names, making the exported BOM and Full Blocks Schedule CSV files easier to understand and use.

23. Preparing Symbols Sheets for Use with BLKEXPORT

If you plan to incorporate BLKEXPORT software into your drawing workflow, you may need to modify your symbols sheet drawings. This typically involves renaming the blocks within the sheet to give each symbol a clear, meaningful name.

Completing this exercise up front is highly beneficial — it eliminates the need to rename blocks for all future drawings. For existing drawings, however, block renaming will likely still be required.

Best Practice from Experience

When I worked as a CAD Technician, I would always copy only the required symbols from the latest symbols sheet into a new drawing. This ensured I was consistently using the most current, approved symbols and saved significant time by avoiding rework or renaming blocks after the drawing was complete.

There is a strong emphasis on the person producing the drawing to exercise due diligence during drawing preparation. Specifically, symbols must be created as blocks where appropriate, rather than drawn as raw geometry.

Critical Requirement

BLKEXPORT will only export objects that are defined as blocks within the drawing. If symbols are not created as blocks, BLKEXPORT cannot export them.

24. Comparing the Summarised BOM with the Source Drawing

Once the drawing is complete and the summarised Bill of Materials (BOM) has been generated, it is good practice to verify the BOM against the symbols in the actual drawing. Print both the drawing and the summarised BOM and compare them.

Traditionally, the BOM is produced manually by printing the drawing, then using a highlighter to count each symbol by hand and record the quantities.

BLKEXPORT eliminates this manual counting process and can generate the BOM substantially faster. However, the accuracy of the summarised BOM depends entirely on the use of blocks in the drawing. There is a strong emphasis on the person producing the drawing to exercise due diligence during drawing preparation — specifically, ensuring that symbols are created as blocks where appropriate, rather than drawn as raw geometry.

BLKEXPORT will only export objects that are defined as blocks in the drawing. If symbols are not created as blocks, BLKEXPORT cannot export them.

25. BOM Accuracy and User Verification

BLKEXPORT generates its results from the block data contained in the supplied DXF file. It does not determine whether a block represents a real-world component, whether a component has been omitted from the drawing, or whether the drawing itself has been prepared correctly.

The accuracy of the resulting BOM therefore depends on the accuracy and structure of the source drawing.

The BLKEXPORT output should be reviewed and verified against the source drawing before being used for procurement, manufacturing, installation, costing, or other decisions where quantity accuracy is critical.

BLKEXPORT is a counting and reporting tool; it does not replace professional review of the source drawing.

26. Application UI Dialog Location on Screen Saved on Exit

The application now remembers the dialog's screen position and restores it the next time it is opened.

27. Symbol Placement and BOM Accuracy

As a rule, symbols belong in model space. You'll still see them in paper space sometimes, but good drawing practice is to keep all symbols in model space.

To get a complete inventory of every symbol in your drawing, use BLKEXPORT and export a full block schedule to CSV. That report lists all symbols present in the dxf drawing file.

Paper-space block insertions are included in BLKEXPORT's block inventory. Users should review these before relying on the summarised BOM, as paper-space blocks may represent legends, title blocks, notes, or other non-physical drawing content.

Why it matters:

- **Model space** = real-world drawing elements that make up the design
- **Paper space** = sheet layout, legends, notes, title blocks
- Symbols in paper space don't represent actual installed components, so they inflate BOM quantities if left in

Quick workflow:

1. Export full block schedule with BLKEXPORT → see all symbols
2. Check for paper space symbols, especially in legends
3. Remove or move them to model space if needed
4. Export the summarised BOM for accurate counts

28. System Stability Requirements for Reliable Software Operation

The BLKEXPORT software can only operate reliably when the underlying operating system is itself stable and functioning correctly.

If your operating system is unstable, unreliable, or behaving erratically, you should ask your IT support team or system analysts to investigate and, if necessary, rebuild or reinstall the operating system on your machine.

As the developer of the BLKEXPORT software, I cannot guarantee reliable or consistent performance of the software when it is running on an operating system that is unstable or malfunctioning. The stability and reliability of the underlying operating system are essential to the proper operation of the software.

29. Block structures that can complicate individuality and should be avoided

Structure	Why it causes problems
Nested blocks	A parent block contains one or more block references. Counting references can count both the parent and its children.
Anonymous blocks (*U###, *E###, etc.)	AutoCAD creates these internally for things like dynamic blocks, arrays, hatches, and other operations. They may not represent a meaningful user-defined block.
Dynamic blocks	Multiple block references can share a common definition but have different visibility states, parameters, stretches, rotations, etc. "Same block" can mean different physical configurations.
Block references of the same definition	Two instances can point to the same block definition but are obviously two separate physical occurrences. You need to distinguish definition identity from instance identity.
Exploded blocks	What was originally one block may now be individual geometry. A block-counting algorithm will miss its former identity entirely.
Blocks containing copied geometry	Users sometimes create visually identical blocks with different definitions/names. Name equality therefore isn't necessarily individuality.

Structure	Why it causes problems
Duplicate/overlapping block references	Two block instances can occupy exactly the same position, potentially looking like one object visually while actually being two database objects.
Blocks inside Xrefs	Objects may appear in the drawing through an external reference rather than being native block references. Counting them as local blocks can produce misleading totals.
Xref block definitions	An Xref itself is represented through block-table structures, so simply iterating block definitions can accidentally count Xrefs as ordinary blocks.
Blocks inside Xrefs inside Xrefs	Recursive external references introduce another form of nesting and ownership hierarchy.
Paper-space vs model-space instances	The same block definition can have independent instances in model space and one or more layouts.
Blocks inside layouts/viewports	Depending on your intended definition of "drawing block," these may or may not belong in the count.
Array-created blocks	Associative arrays can generate multiple occurrences from one underlying operation/structure. Treating the array definition as one block can differ from counting its resulting instances.
Visibility-state variants	Dynamic blocks can have geometry hidden/shown depending on state. Two instances with the same definition may have materially different visible content.
Mirrored blocks	Negative scaling can produce a geometrically transformed instance that is still the same block definition.
Non-uniformly scaled blocks	Identical definitions can have very different physical sizes, making geometry-based individuality tricky.
Rotated/transformed instances	Position, rotation, scale and potentially UCS transformations need to be considered if you're identifying physical occurrences.
Blocks containing attributes	Attribute references belong to a particular block instance. They shouldn't be independently counted as blocks.
Blocks containing proxy objects/custom objects	Vertical products such as Civil 3D/Architecture can introduce objects whose behaviour isn't equivalent to ordinary AutoCAD entities.
Overlays / multiple references to the same Xref	The same external content may appear multiple times through different references.
Temporary/generated blocks	Some AutoCAD commands/features generate implementation blocks that aren't necessarily meaningful objects from the user's perspective.

30. Block Counting Definition

BLKEXPORT counts **individual top-level block instances** in model space and paper space only.

A block is counted when it represents an independent block occurrence in the drawing. **Block references contained within another counted block are not counted separately**, preventing nested blocks from being double-counted.

BLKEXPORT can count **both named and anonymous block references**, including AutoCAD anonymous blocks such as *U###, where they represent an independent top-level block instance.

By default, BLKEXPORT does not independently count:

- **Nested block references** contained within another counted block
- **Block references belonging to an external reference (Xref)**, unless specifically configured otherwise
- Other block references or structures that are **not independent top-level occurrences** in model space or paper space

The objective is to ensure that each reported block represents an **individual occurrence at the top level of the drawing**, rather than simply counting every block reference present anywhere in the drawing database.

Important: A block definition and a block instance are not the same thing. Multiple individual block instances may share the same block definition, and each independent instance is counted separately by BLKEXPORT.

31.DXF Conversion Important!

When a drawing has been converted from DWG to DXF, users should verify the resulting DXF before relying on the exported data.

BLKEXPORT processes the contents of the DXF file supplied to it. It cannot verify whether information, block structures, attributes, external references, or other drawing data were altered or omitted during conversion.

Where the drawing has been converted to DXF, the exported results should therefore be checked against the source drawing.

32.External References (Xrefs)

BLKEXPORT does not treat block references contained within external references (Xrefs) as independent blocks in the host drawing.

If a drawing relies on Xrefs for symbols, equipment, annotation, or other components, those objects may therefore not appear in the BLKEXPORT results for the host DXF.

Users should identify and review any Xrefs used by the source drawing when determining whether the exported BOM represents the complete design.

33.Dynamic & Smart Blocks

Warning: Dynamic Blocks and Smart Blocks are not supported in BLKEXPORT v1.0. They are excluded from block counts and CSV exports. Drawings containing these objects may therefore produce incomplete BOMs.

